SAMARTH

CUET PG 2027

Lecture - 10 | Computer Science

Unit 03 : Digital Fundamental

Topic : Kmap Simplification: Part 1

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Recap *of previous lecture*

1. Boolean Algebra and Logic Gates: Part 2



Topics *to be covered*

1. Kmap Simplification: Part 1

↳ Solving Boolean Exp but without using laws



Topic : Kmap Simplification: Part 1

$= 2^n$
$= 2^2$
$= 4$

Expression

Sum of Product (SOP) (minterm)

Product of Sum (POS) (maxterm)

Ex: x, y

Possible minterms :-

xy
$x'y$
xy'
$x'y'$

SOP :

$xy + x'y + xy' + x'y'$

Ex: x, y

Possible Maxterm.

$(x+y)$
$(x'+y)$
$(x+y')$
$(x'+y')$

POS $\Rightarrow (x+y)(x'+y)(x+y')(x'+y')$

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Minterm-

- A minterm of variables is A product of a literals in which each variable appears exactly once in either true or complimented form, but not both
  For example, the list of all of the minterms of the two variables x and y are:
  xy, x'y, xy', x'y'
  xyz, xyz', xy'z, x'yz, xy'z', x'y'z, x'yz', x'y'z'
  in a similar way, n variables can be combined to form 2^n minterms.

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Maxterm-

- A maxterm of n variable is a sum of n literals in which each variable appears exactly once in either true or complicated form, but not both.
  For example, all maxterm of two variables x and y are:
  $x+y, x'+y, x+y', x'+y'$
  And all max terms of three variables x, y and z are
  $x'+y'+z', x'+y'+z, x'+y+z', x+y'+z', x+y'+z, x+y+z', x'+y+z, x+y+z$
- In a similar manner, n variables forming maxterms, with each variable being complemented or un-complentented, provide 2^n possible combination.

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- Note: Any single minterm will be zero except for a single combination of the values of the literals while any single maxterm will be 1 except for a particular combination of the values of the literals. For instances, xy'z will be 0 for any combinations of x, y and z except x = 1, y = 0 and z= 1. Also, x+y'+z will be I for any combination for x, y and z except for x = 0, y = 1 and z = 0.

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- When a Boolean function is written as a sum of minterms, it is referred to as a minterm expansions or the disjunctive normal form of the Boolean function. It is also called canonical sum of products or standard sum of products. The following are minterm expansions of Boolean functions
  1. $f(x, y) = xy + x'y$
  2. $f(x, y, z) = x'y'z + x'yz + xy'z' + xyz$

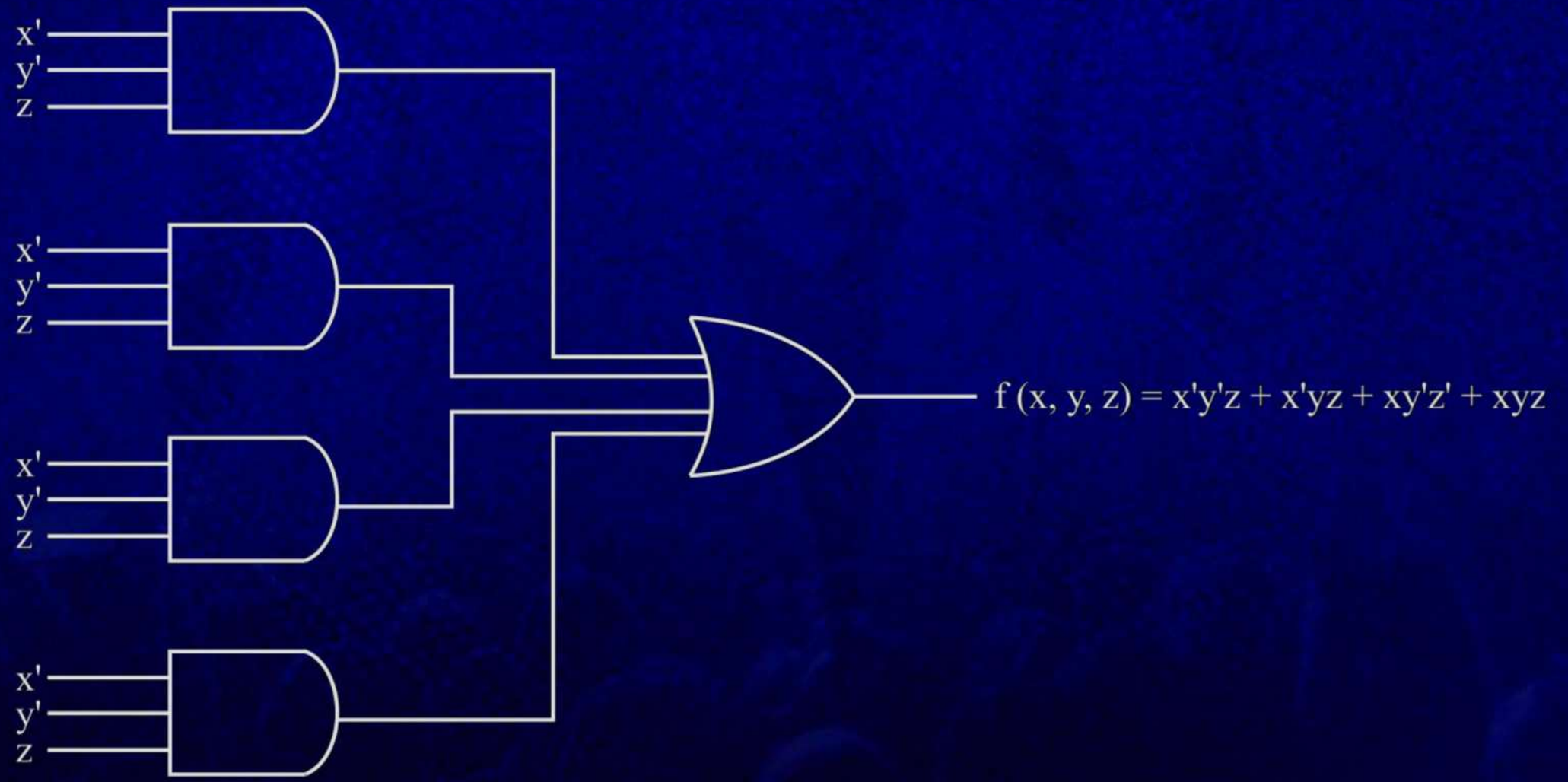
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- The canonical sum of products form can be implemented simply by an interconnection of gates. Note that each mean tern is realised using single AND gate. The outputs of all the minterms are connected to the inputs of a single OR gate whose output is the desired function. Therefore, if there are n_1, minterms in the canonical sum of products, there will be n_1+1 gates used in its realisation.

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- When a Boolean function f is written as a product of maxterm, it is referred to as a maxterm expansion or the conjunctive normal form of the Boolean function. It is also called canonical product of sums and standard product of sums. The following are maxterm expansions of Boolean functions
  1. $f(x, y) = (x+y)\,(x'y)$
  2. $f(x, y, z) = (x+y+z)\,(x+y'+z)\,(x'+y+z')\,(x'+y'+z)$

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Topic : Kmap Simplification: Part 1

- Actually either canonical form may use many more gates than is necessary. In later section we shall discuss a procedure for minimizing the number of gates required.
- Boolean functions expressed as a sum of minterms or product of maxterms are said to be in canonical form.
- A Boolean function when expressed as a sum of all 2" minterms of n variables is called the complete minterm expansion or the complete disjunctive normal form. Similarly, a Boolean function when expressed as a product of all 2" maxterms of n variables is called the complete maxterm expansion or complete conjunctive normal form.

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- Boolean function when expressed as a complete minterm expansion or complete maxterm expansion form is called complete canonical form
- The complete minterm expansion in 'two variables x and y is
  xy + xy′ + x′y + x′y′
- Which can be proved identically equal 1 as shown below:
  xy + x'y + xy' + x'y' = xy + xy′ + x'y + x'y′
  = x (y +y′) +x' (y + y′)
  x.1 + x'.1
  = x + x′
  = 1

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- In general the complete minterm expansion form in n variable is identically 1.

Expansion of a Boolean function as a canonical form-

- A Boolean function may be expressed to minterm or maxterm expansion either using a truth table or algebraically.

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Form a given truth table-

- If a truth table to describe the behaviour of logic network is given, the corresponding logic expression in the minterm expansions and maxterm expansions can be obtained from the truth table by the method described below.
  (a) Inspect the column corresponding to dependent variable from the top row a 4 pick the row with 1 entry.
  (b) A term in the Boolean expression corresponding to this row is obtained ANDing all the independent variables in the truth table.
  (c) Repeat steps (a) and (b) till all the entries in the dependent variable column is exhausted. Obtain a Boolean expression by ORing the terms corresponding to 1 entries of dependent variable.

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- The method is applied for obtaining minterm expansion of a Boolean function. The method for obtaining the maxterm expansion of a Boolean 1 similar, instead of picking the rows of 1 entry in dependent variable, the rows of O entry are to be picked up. The individual term is formed by taking the sum of independent variables. The expression is obtained by ANDing the individual terms. The following example clarifies this procedure.

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Minterm – Consider O/P 1 ⇒ Input 0 Complement
Input 1 without 11

$x'y'z' + x'y'z + x'yz' + xy'z' + xy'z$.

Q Expected.

- **Example:** Consider the Following truth table and find the corresponding Min-term and Max-term:

→ Consider O/P 0 ⇒ Input 0 without complement
Input 1 Complement

$(x+y'+z')(x'+y'+z)(x'+y'+z')$

	Input			Output
	X	Y	Z	F
$x'y'z'$	0 x'	0 y'	0 z'	1
$x'y'z$	0 x'	0 y'	1 z	1
$x'yz'$	0 x'	1 y	0 z'	1
$(x+y'+z')$	0 x	1 y'	1 z'	0
$xy'z'$	1 x	0 y'	0 z'	1
$xy'z$	1 x	0 y'	1 z	1
$(x'+y'+z)$	1 x'	1 y'	0 z	0
$(x'+y'+z')$	1 x'	1 y'	1 z'	0

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- Solution: Corresponding to entry 1 of the output column in the first, second, third, fifth and sixth row the minterms are X'Y'Z', X'Y'Z, X'YZ', XY'Z', XY'Z. Hence the minterm expansion of the Boolean function given in truth table is:

  F = X'Y'Z' + X'Y'Z + X'YZ' + XY'Z' + XY'Z

  Corresponding to 0 entries of the output column in the fourth, seventh and eighth row the maxterms are X + Y' + Z', X' + Y' + Z and X' + Y' + Z'. Hence the maxterm expansion of the Boolean function given in truth table is:

  F = (X + Y' + Z')(X' + Y' + Z)(X' + Y' + Z')

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Algebraic method-

- To obtain the minterm expression by algebraic method, first write the expression as a sum of products and then introduce the missing variable in each term by applying the theorem a + a = '1. For finding maxterm expansion, factor the function to obtain a product of sums, introduce the missing variable in each sum term by using aa'=0 and then factor again to obtain the maxterms.

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$x + x' = 1$

- **Example:** Express the Boolean function $f(x, y, z) = x + y' z$ in a sum of minterms.

$x + y'z$ (1) (2)

Solution: The function has three variables x, y and z.

The first term x is missing two variables; therefore applying a + a =1

$x = x(y + y') = xy + xy'$

This is still missing the variables z:

$x = xy(z + z') + xy'(z + z')$

$= xyz + xyz' + xy'z + xy'z'$

$xy'z + x'y'z + xyz + xyz' + xy'z'$

The second term y'z is missing one variable x:

$y'z = y'z(x + x') = xy'z + x'y'z$

Hence $f(x, y, z) = x + y'z$

$= xyz + xyz' + xy'z + xy'z' + xy'z + x'y'z$ using (1) and (2)

$= xyz + xyz' + xy'z + xy'z' + x'y'z$ applying the theorem

a + a = a on xy'z

(2) $y'z(x + x')$

Combining:

$= xy'z + x'y'z$

(1) $x(y + y')$

$= xy + xy'$

$= xy(z + z') + xy'(z + z')$

$= xyz + xyz' + xy'z + xy'z'$

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ab + a'c Not in maxterms
use Distributive law
(ab + a')(ab + c) ⇒ (a + a')(b + a')(a + c)(b + c)
① ② ③

- **Example:** Express the Boolean function f(a, b, c) = ab + a'c as a product of maxterms.
- **Solution:** First convert the function into factor to obtain = the product of sum forms by using distributive law x +y, z =(x+ y) (x + z)

a + a' = 1

Cc' = 0

f(a, b, c) = ab + a'c = (ab + a') (ab + c)
= (a + a) (b +a) (a +c) (b+c)
= (b +a') (a + c) (b +c) (a + a' = 1)

① b + a' + cc'
= (a' + b + c)(a' + b + c')

Since each sum term is missing one variable, applying aa' = 0 we get
a' + b = a' + b + cc' = (a' + b + c) (a' + b +c')
a + c = a + c + bb' = (a + b + c) (a + b' + c)
b + c = b + c + aa' = (a + b + c)(a' + b + c)

② a + c + bb'
(a + b + c)(a + b' + c)

Now removing those terms that paper more than once (a, a =a),
f(a, b, c) = (a + b + c) (a + b' +c) (a' + b +c) (a' + b +c')
which is a product of maxterms.

③ b + c + aa'
(a + b + c)(a' + b + c)

(a' + b + c)(a' + b + c')(a + b + c)(a + b' + c) Ans

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- The Karnaugh map method is a graphical technique which provides a simple straightforward procedure for simplification of Boolean expressions of two, three or four variables, it can also be extended to functions of five, six, or more variables. But as the number of variables increases, the excessive number of squares prevents a reasonable selection of adjacent squares and it is difficult to be sure that the best selection has been made.

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- A Karnaugh map (K-map is a diagram made up of a number of squares. If the expression contains n variables, the map will have 2" squares. Each square represents a minterm and Is are written in the corresponding squares for the minterms present in the expression and Os are written in those squares which correspond to the minterms not present in the expression. Once the map is filled with Os and 1s, the canonical sum of products expression for the output can be obtained by ORing together those squares that contain 1. It is a two dimensional representation of a truth table. as it can display in a visual form all the information contained in a truth table.

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2-variable K-Map:

- Since the number of variables are 2, the map will have 22 = 4 squares. The values of one variable, say A, are listed above the top horizontal line and the values of other variable, say B, are listed on the left side. Four possible min terms with two variables A and B.
- AB, AB', A'B, A'B' are represented by the four squares in the map and alternative representation in figure. Squares are said to be adjacent if the minterms that they represent differ in exactly one literal. For instance the square representing A B is adjacent to the squares representing AB and A'B'.

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A	B
0	0
0	1
1	0
1	1

no of squares $= 2^n$
$= 2^2$
$= 4$

B \ A	A'	A
B'	A'B'	AB'
B	A'B	AB

B \ A	0	1
0	00 $\bar{A}\bar{B}$	01 $\bar{A}B$
1	10 $A\bar{B}$	11 AB

B \ A	0	1
0	A'B'	AB'
1	A'B	AB

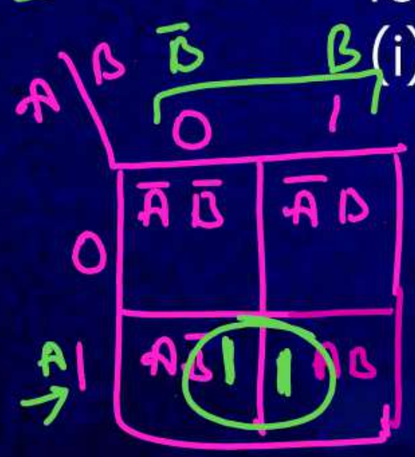
By default SOP ⇒ 0 – complement
1 – without ||

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- The expression can be simplified by properly combining those squares in the K-map which contains 1s. The process for combining 1s is called looping. Whenever there are Is in two adjacent squares in the K-map, the minterms represented by these squares can be looped and it eliminates the variable that appear in complemented and uncomplemented form.

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- **Example:** Find Karnaugh maps and simplify the expressions (i) AB' + A'B' (ii) AB' + A'B (iii) AB'+ A'B+ A'B'
- **Solution:** Since each expression contains two variables, three K-maps of four squares are shown below:



(i) two adjacent squares A'B' and AB' containing 1 have been grouped together. They have been encircled. These two terms can be looped that eliminates the A variable since it appears both in complemented and uncomplemented forms. This can be verified algebraically as follows:

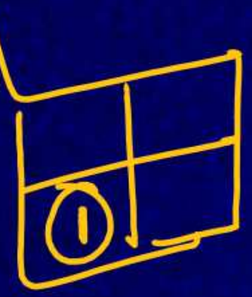
AB + A'B'= (A+A')B
= 1.B' = B'





(ii) K-map does not contain any adjacent square of minterms containing 1, hence the expression can not be simplified further.

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B \ A	0	1
0	1	1
1	0	0

B \ A	$\bar{A}$ 0	A 1
$\bar{B}$ 0	0	1
B 1	1	0

Already solved.

$\bar{A}B + A\bar{B}$ ✓

Q $AB' + A'B + A'B'$

B \ A	$\bar{A}$ 0	A 1
$\bar{B}$ 0	1	1
B 1	1	0

$\bar{B}$

$\bar{A}$

$\bar{A} + \bar{B}$

Ans ✓

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(III) These are two pairs of 1s is and they can be combined .Notice that 1 in first column and first row has been enclosed twice, as it is permissible to use the same 1 more than once. Looping of horizontal 1-squares gives the result B' and vertical 1-squares gives A. Hence the given expression reduces to the simplified form as A'+B' This can also be verified algebraically as follows:

$AB'\ A'B + A'B' = AB' + A'\ (B + B')$

$= AB' + A$

$= A'+B'$

3 variables (let A B C)

No of squares $= 2^3$
$= 8$

① C \ AB

	00	01	11	10
0	0	2	6	4
1	1	3	7	5

② AB \ C

	0	1
00	0	1
01	2	3
11	6	7
10	4	5

③

A \ BC

	00	01	11	10
0	0	1	3	2
1	4	5	7	6

Pair of 1's

① Pair of 4 1's (Quad)
② Pair of 2 1's (Double)
③ Single 1

Horizontal ✓
Vertical ✓
Diagonal ✗

2 Mins Summary

SOP
POS
} + Expected O/P

Keep into
2 variable

3 variable introduce

Homework from SAMARTH CUET PG 2027 Module

Yt video

By Puneet Kumar

BCD codes

Excess 3 codes, Gray codes

THANK YOU